Ahmedabad University

CSE400

CLIMATE AND WEATHER SYSTEMS

Problem Statement: *Probabilistic Precipitation Nowcasting*

Section 2 - Group 2

Dhruvi Faldu (AU2440216)
Shine Shah (AU2440148)
Vamsi Pancholi (AU2440023)
Sharanya Gupta (AU2440048)
Rim (AU2440262)

Project Background and Motivation

- This project targets the extension of a deterministic rainfall nowcasting system into a probabilistic system. The traditional deterministic models produce a single forecast using fixed inputs and assumptions, thus limiting them when it comes to highly dynamic and uncertain systems such as weather. Randomness is added to the main aspects of the system making it an ensemble model that yields more than one possible scenario of the future.

- In the real world, precise rainfall forecasting is extremely important to applications such as flood forecasting, disaster management, agriculture, and urban planning. Nevertheless, the making of decisions in these sectors is not based on precise values but on risk and probabilities.

- A deterministic forecast is not able to capture the extent of confidence of the system like a probabilistic forecast that can be used to give information on the probability of the occurrence of heavy rainfalls. This could help to make the system more practical in making decisions.

- All in all, what makes this issue significant is the fact that weather systems are uncertain in their nature. Deterministic models are usually overconfident, and do not respond to sudden changes, whereas probabilistic models are a more realistic model since they provide a model of natural variability and the lack of knowledge.

Base Paper Overview – Pysteps Framework

Pysteps: an open-source Python library for probabilistic precipitation nowcasting (v1.0)

Main Idea

- Perform short-term rainfall forecasting (0–6 hours)
- Use radar data + motion estimation
- Generate probabilistic (ensemble) forecasts

Key Contributions

- Combines:
- Deterministic advection (motion-based prediction)
- Stochastic perturbations (uncertainty modelling)
- Introduces ensemble nowcasting
- Uses cascade decomposition for multi-scale rainfall modeling

Core Model

$$R_t = \sum \phi_k R_{t-k} + \epsilon_t$$

- Deterministic part → past rainfall
- Stochastic part → uncertainty

Why It Is Important?

- Captures uncertainty in rainfall evolution
- Provides probability-based forecasts
- Widely used in weather and flood prediction systems

Problem Formulation

- Rainfall is hard to predict and is constantly changing, so a single forecast cannot capture all possible outcomes.

- Nowcasting is a way to predict rainfall that is going to happen like in the next 0 to 6 hours. It uses radar data to make the predictions faster and more up to date than traditional weather forecasting

- Uncertainty comes due to two reasons- errors in reading current radar data, and the fact that storms grow, decay, and change direction in a way we cannot predict.

- The challenge is to communicate this uncertainty in a useful way helping decision makers like flood managers and city planners act in time during extreme weather

- To handle this, multiple possible future rainfall scenarios are generated instead of just one, forming an ensemble forecast

Mathematical Model

- Rainfall intensity is always positive and can be extremely large during storms, so it is modeled using a log-normal distribution, which handles this skewed behaviour well.

- Storm movement is uncertain and can change suddenly so it is modeled using a Laplace distribution, which has heavier tails than a normal distribution and better captures large unexpected shifts.

- The Bernoulli model is used to check the exceedance of rainfall from a certain threshold point, clearly indicating whether the exceedance is occurring or not at each instance.

- Markov processes are utilized to capture the sequential evolution of rainfall intensity levels along with the embedded stochastic noise in the system.

- Jensen’s inequality is applied during the log transformation of rainfall data to account for convexity bias in the expectation.

- Chebyshev’s inequality is used to quantify how reliable the estimated probability of rainfall exceedance truly is, without assuming a specific distribution.

Bounds Used in the Problem

1. Markov's Inequality - $P(R \geq r) \leq E[R]/r$ → *Used for extreme rainfall risk*

2. Chebyshev's Inequality - $P(|R - E[R]| \geq k) \leq Var(R)/k^2$ → *Used for deviation from mean*

3. Jensen's Inequality - $E[e^Y] \geq e^{E[Y]}$ → *Used for log transformation analysis*

4. Constraints

- Rainfall $R \geq 0$
- Log rainfall -> Normal distribution
- Motion uncertainty -> Laplace distribution

Why Needed?

- Handle uncertainty without exact distributions
- Provide safe upper bounds for extreme events
- Support decision-making (flood risk)

Deterministic S-Prog Nowcasting

Input:
- Radar rainfall: R(t−2),R(t−1),R(t)

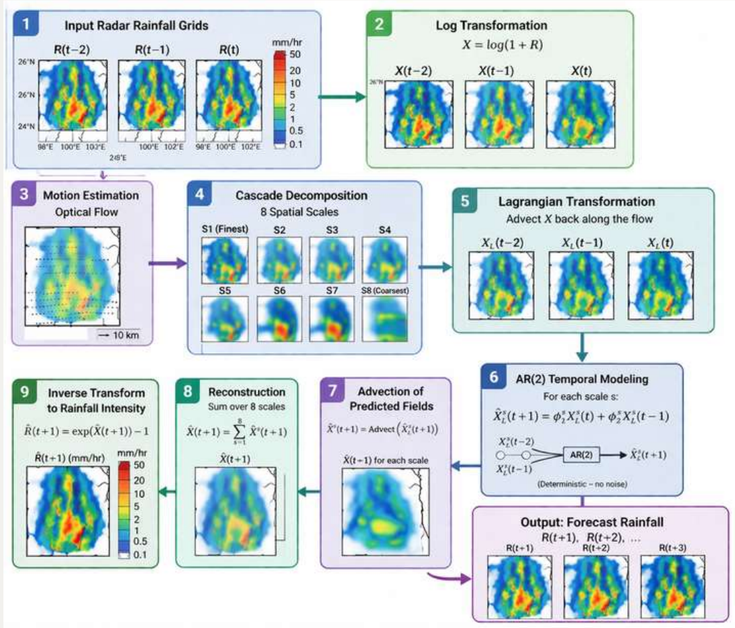
Processing Steps:
- Log transformation
- Motion estimation
- Cascade decomposition

Core Model (AR(2)):
- R(t) = ϕ1 R(t−1) + ϕ2 R(t−2)
- Uses past rainfall to predict future

Deterministic Nature
- No randomness (no noise added)
- Same input → same output

Final Steps
- Advection (movement of rain)
- Reconstruction

Output
- Future rainfall prediction
- Threshold exceedance detection



Randomized Approach

Why Randomization?

- Rainfall is highly uncertain and dynamic. The deterministic model produces only a single forecast and cannot capture variability in rainfall evolution. To address this, randomness is introduced to generate multiple possible future scenarios.

How Randomness is Introduced

• Gaussian Noise in AR(2) Model

$$R(t) = \phi_1 R(t-1) + \phi_2 R(t-2) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Models uncertainty in rainfall growth and decay.

• Laplace Motion Perturbation

$$V' = V + \eta, \quad \eta \sim \text{Laplace}(0, b)$$

Models uncertainty in storm movement and direction.

Ensemble Forecasting

- The model is executed multiple times (e.g., 20–30 simulations), each with different random noise, producing a set of forecasts:

{R1,R2,...,RN}

Probabilistic Outputs

• Mean rainfall forecast

• Uncertainty (ensemble spread)

• Probability of exceedance:

P(R>threshold)

CODING AND SIMULATION

Ensemble Nowcasting with Laplace Motion Perturbation

- Short-term rainfall forecasting
- Compare:
- Deterministic vs Ensemble

- Goal: Improve accuracy + quantify uncertainty
- Libraries used: PySTEPS, NumPy, SciPy, and Matplotlib

Implementation Overview

Radar Frames
FMI PGM
dBZ → Rain Rate
→ Log

Motion Field
Lucas-Kanade
Optical Flow
(last 3 train frames)

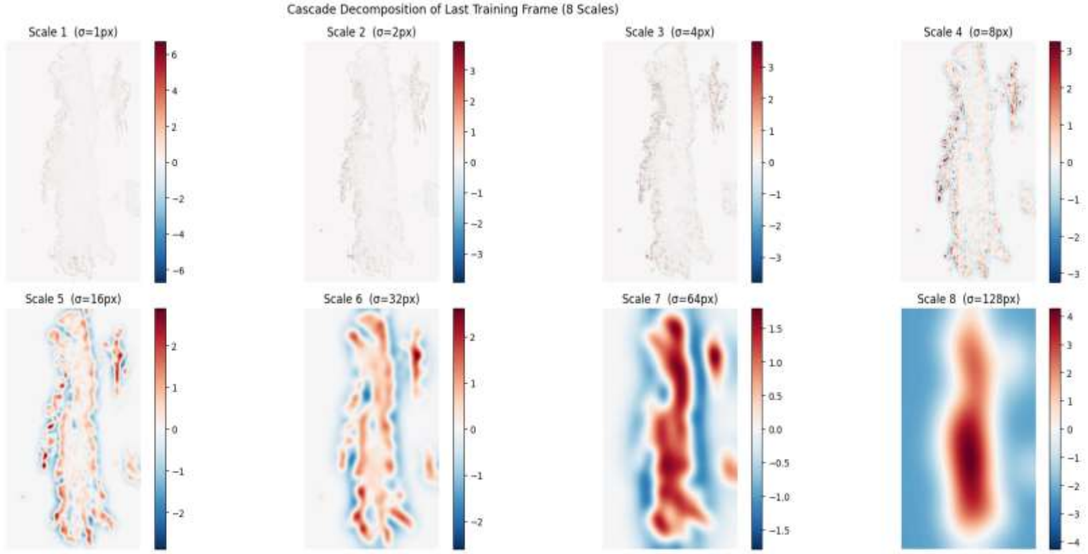
Cascade Decomp
8 Gaussian
Spatial Scales
(1px → 128px)

AR(2) Model
Per-scale
ϕ_1, ϕ_2, σ
(train only)

Ensemble Forecast
30 members
± Laplace MP
AR(2) noise

Strict Train/Test Split	Future frames never seen during training; evaluation uses held-out test set only
Calibrated Noise	AR(2) σ estimated from actual training residuals, not a free parameter
Laplace Motion Perturbation	Heavy-tailed noise on motion field; spatially smoothed & decay-weighted by lead time
Evaluation	MSE, RMSE, MAE, Bias + Spread-Skill ratio across 6 lead times (5m – 30m)

Simulation Setup — Cascade & Motion Uncertainty



Scales 1–4

Fine pixel-level detail
(1px → 8px σ)
Sharp rain cell edges

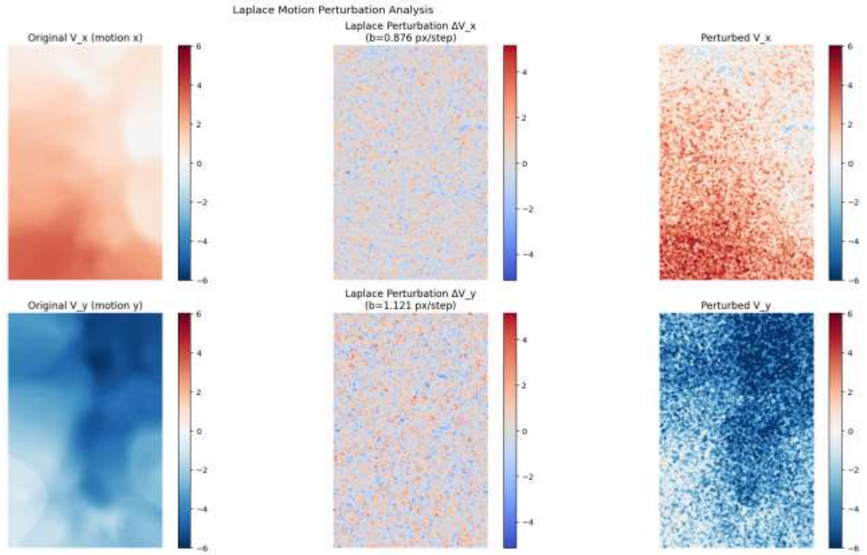
Scales 5–8

Mesoscale storm structure
(16px → 128px σ)
Broad envelope

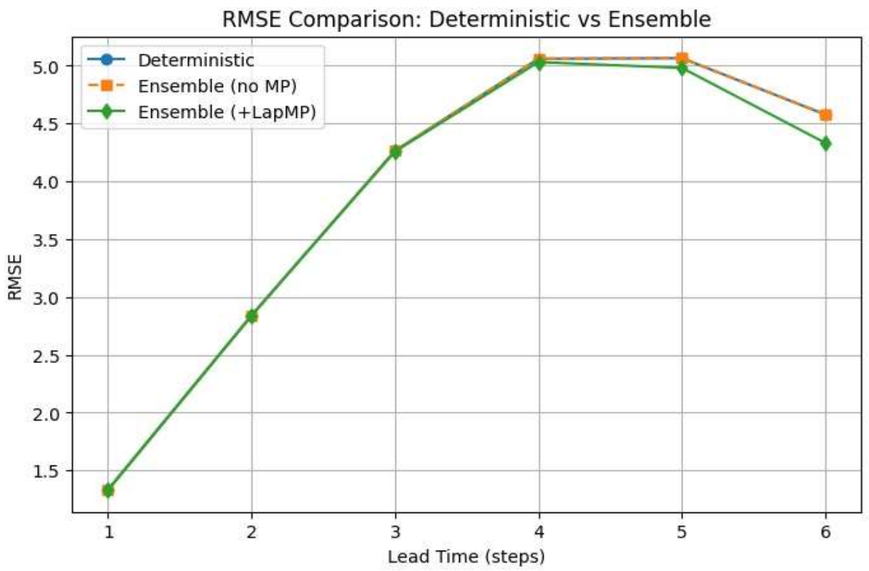
Why Laplace?

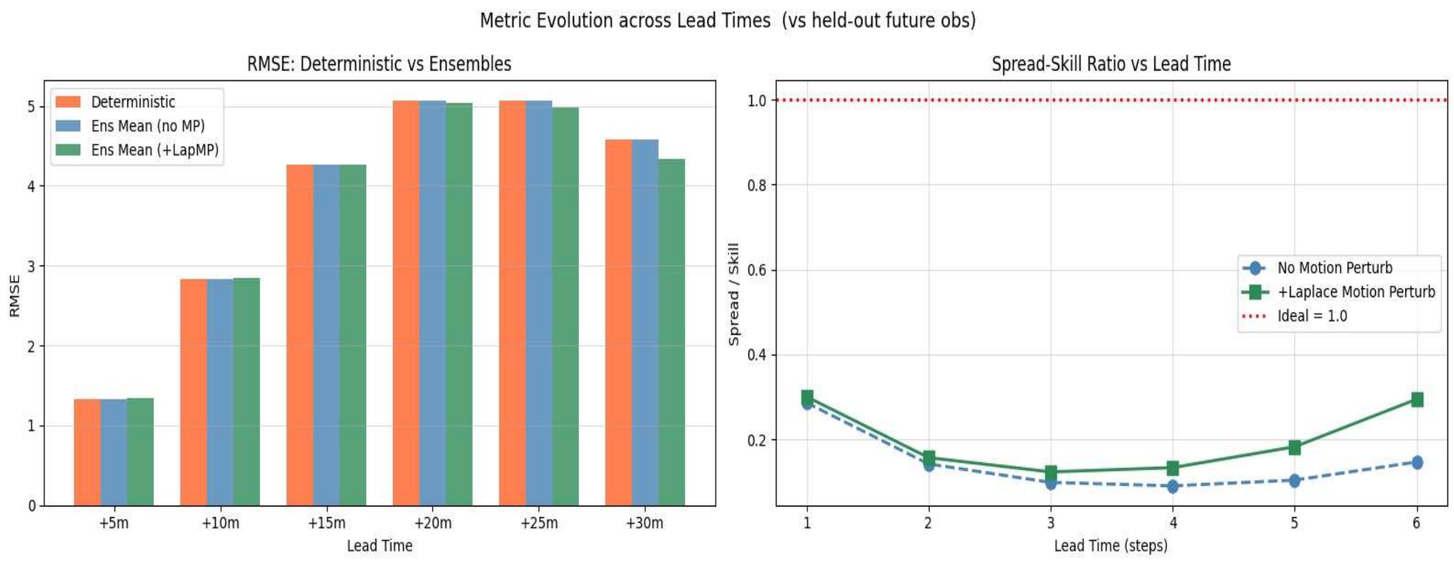
- Laplace allows **extreme variations**
- Better than Gaussian for weather systems

- Small **b** → small noise → little change in motion
- Large **b** → large noise → more uncertainty



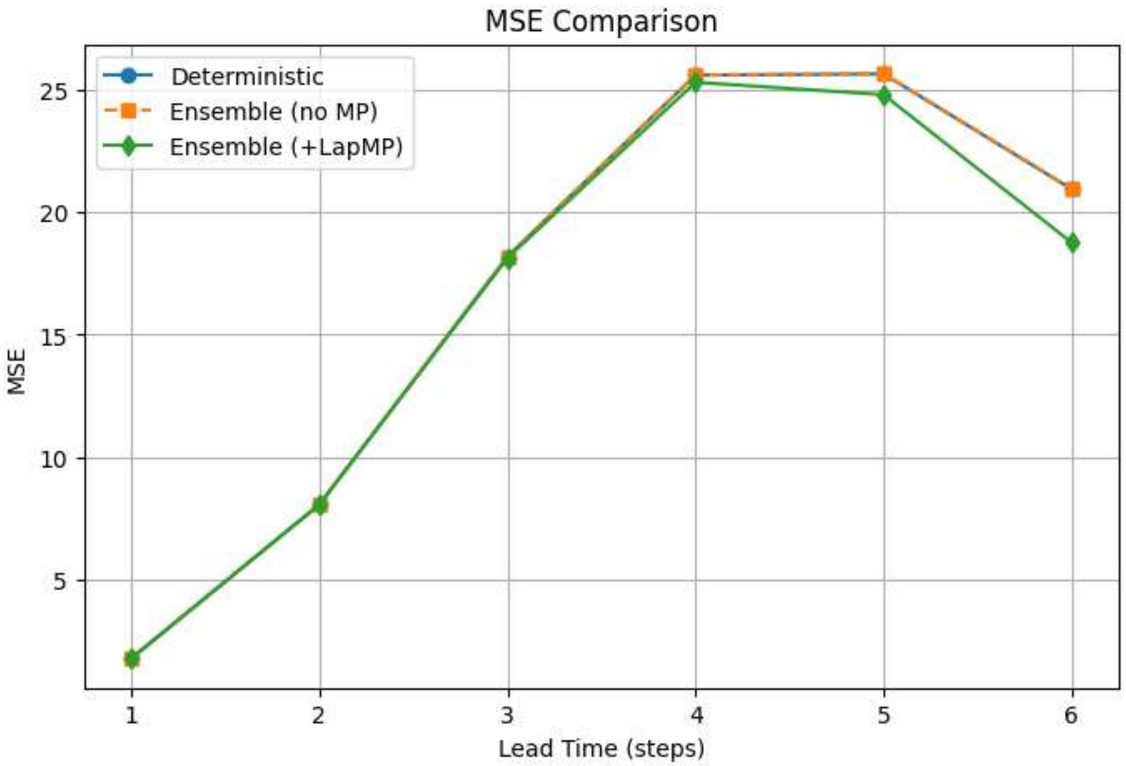
Results — RMSE, MAE, MSE & Spread-Skill

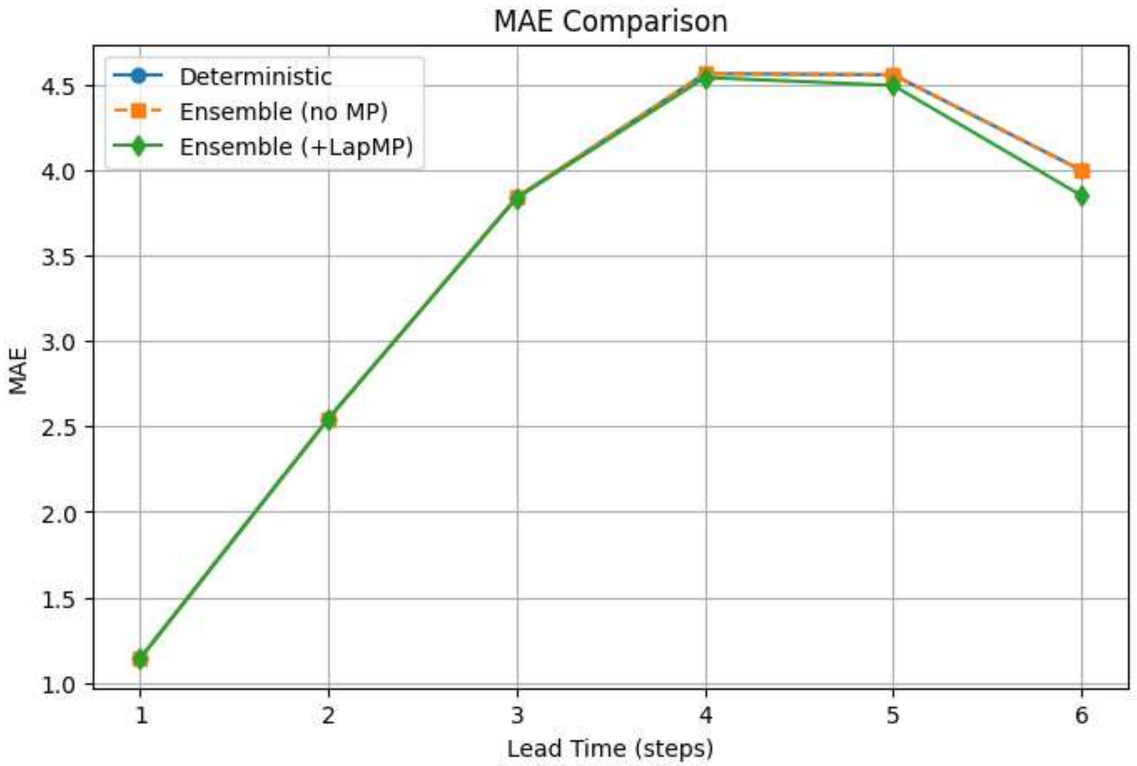




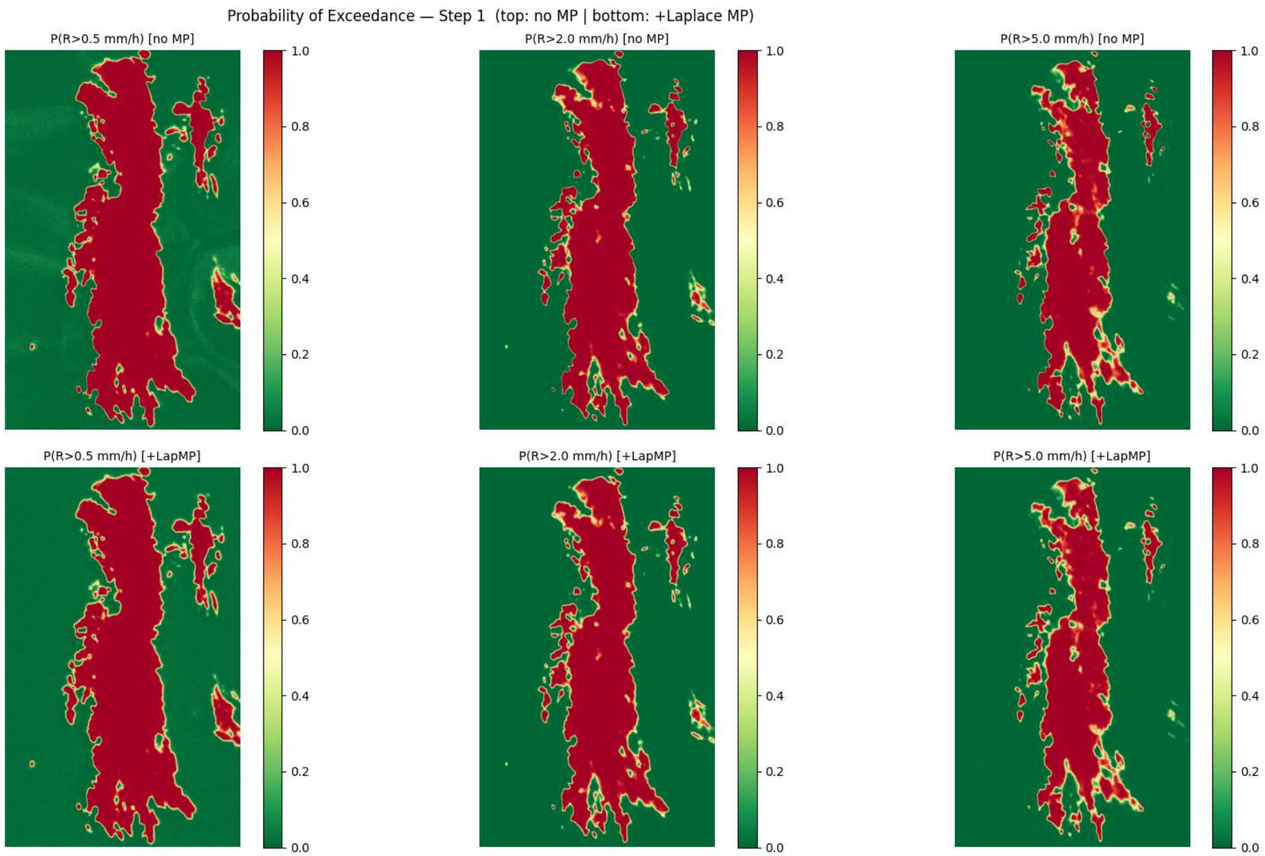
≈1.35	≈5.0	+30%	30
Step-1 RMSE (all methods)	Peak RMSE at +20 min	Spread-Skill gain with LapMP	Ensemble members

Results — RMSE, MAE, MSE & Spread-Skill





Probabilistic Output & Key Takeaways



Calibrated boundaries

Laplace MP produces softer, more graduated probability contours at high thresholds

Spread-Skill improves

LapMP ratio closer to ideal = 1.0 at later lead times vs without perturbation

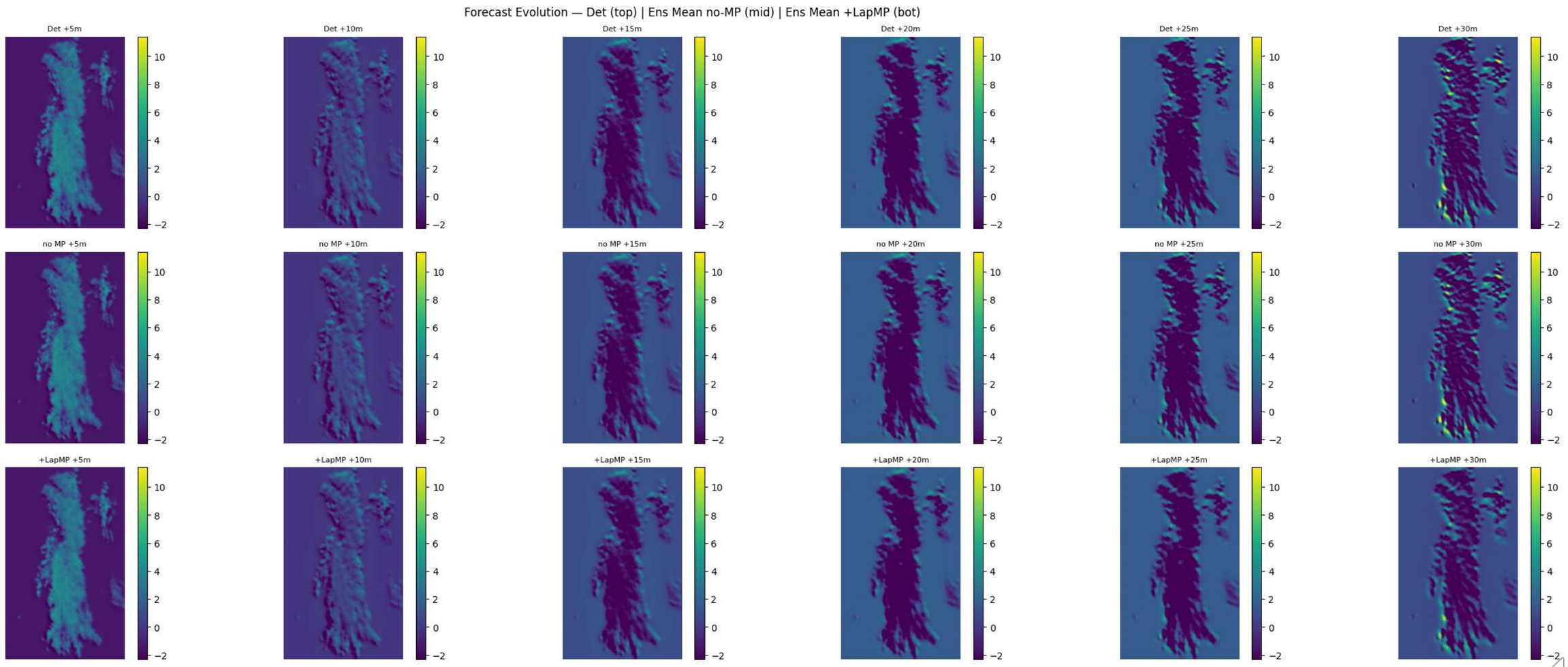
No accuracy penalty

Ensemble mean RMSE matches deterministic — added spread costs nothing in skill

Future work

Dynamic Laplace scale, larger ensembles, longer radar sequences, probabilistic verification (CRPS)

Forecasting Results



Inference - Domain Perspective

- From a domain perspective, the project shows that predicting rainfall involves various types of uncertainty. This includes natural randomness in atmospheric processes and uncertainty from imperfect knowledge of motion and measurements.

- Different kinds of randomness need separate modeling. For instance, small changes in rainfall intensity can use Gaussian noise, while sudden shifts in motion require heavier-tailed distributions.

- The results indicate that motion uncertainty is important for making forecasts more realistic.

- While deterministic models may be slightly more accurate in some cases, probabilistic ensemble models offer extra benefits, like uncertainty estimation, probability maps, and better reliability. This is especially true for complex and rapidly changing weather conditions.

Inference - Computer Science Perspective

- From a computer science perspective, the project proves that randomized algorithms are effective in modeling real-world systems.

- By adding controlled randomness to the prediction and motion parts, the system shifts from a strict deterministic approach to a flexible probabilistic one. Using ensembles is similar to a Monte Carlo simulation method, where several runs help approximate the distribution of possible outcomes.

- The project also points out key trade-offs, like higher computational costs versus greater robustness and interpretability. Additionally, it highlights the need for better evaluation metrics that measure not just accuracy but also the quality of uncertainty estimates.

- Overall, the system shows that blending deterministic pipelines with stochastic modeling results in more reliable and practical solutions.

References

- Pysteps: an open-source Python library for probabilistic precipitation nowcasting (v1.0)
- https://www.metoffice.gov.uk/research/weather/ensemble-forecasting/what-is-an-ensemble-forecast
- https://en.wikipedia.org/wiki/Ensemble_forecasting

Ahmedabad University

THANK YOU!